

Module 14: Advanced Visualizations in Power BI

Overview

Basic charts such as bar charts, line charts, and pie charts are useful for many reporting scenarios. However, modern business intelligence often requires deeper analysis and more sophisticated visual storytelling.

Power BI provides several advanced visualizations that help users analyze trends, identify root causes, understand customer behavior, track conversion processes, and discover hidden insights.

This module covers advanced charts, AI-powered visuals, analytical reporting techniques, and real-world use cases.

Learning Objectives

After completing this module, you will be able to:

- Understand advanced Power BI visualizations
 - Use Waterfall Charts effectively
 - Analyze sales pipelines using Funnel Charts
 - Explore relationships with Scatter Charts
 - Visualize hierarchies using Treemaps
 - Monitor KPIs using Gauges
 - Compare rankings using Ribbon Charts
 - Perform root-cause analysis with Decomposition Trees
 - Use AI-powered visuals
 - Select appropriate advanced visuals for business scenarios
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Table of Contents

1. Introduction to Advanced Visualizations
2. When to Use Advanced Visuals
3. Waterfall Charts
4. Funnel Charts
5. Scatter Charts
6. Bubble Charts
7. Treemaps
8. Gauges
9. Ribbon Charts
10. Decomposition Tree
11. Key Influencers Visual
12. Q&A Visual
13. Smart Narrative

14. Custom Visuals
15. Analytical Reporting Techniques
16. Choosing the Right Advanced Visual
17. Best Practices
18. Common Mistakes
19. Key Terminologies
20. Practice Exercises
21. Mini Project
22. Module Summary

1 Introduction to Advanced Visualizations

Advanced visuals go beyond basic reporting.

They help answer complex questions such as:

Why did revenue decline?

Which factor influences customer churn?

Where are conversion losses occurring?

Which product categories drive profit growth?

Why Advanced Visuals Matter

They provide:

- Deeper Insights
- Better Storytelling
- Root Cause Analysis
- Decision Support

2 When to Use Advanced Visuals

Use advanced visuals when:

Traditional Charts Are Insufficient

Example:

A bar chart may show declining revenue but not explain why.

You Need Root Cause Analysis

Use:

- Decomposition Tree
 - Key Influencers
-

You Need Process Analysis

Use:

- Funnel Chart
-

You Need Variance Analysis

Use:

- Waterfall Chart
-

3 Waterfall Charts

Purpose

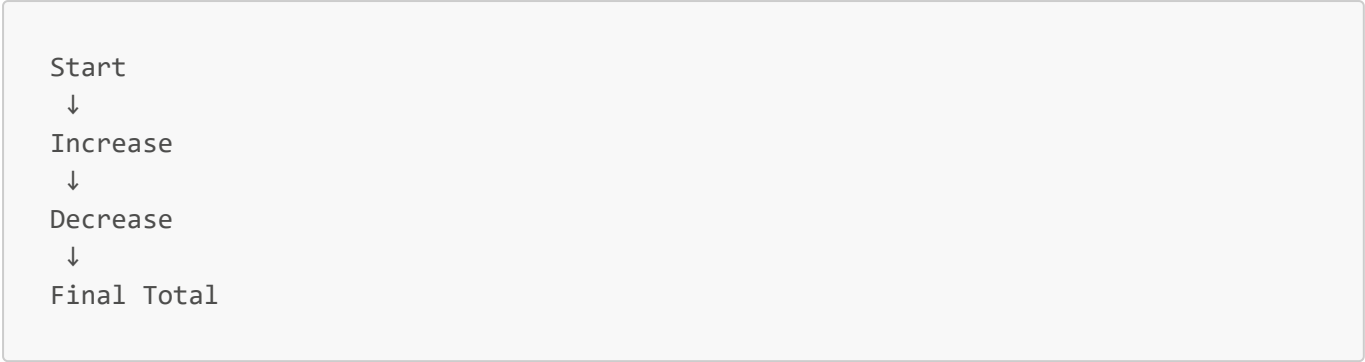
Shows how individual components contribute to a final value.

Example

Revenue Breakdown

```
Starting Revenue
+ Product Sales
+ Service Revenue
- Discounts
- Returns
-----
Final Revenue
```

Visualization Concept



Business Use Cases

Revenue Analysis

Understand revenue drivers.

Profit Analysis

Track profit changes.

Budget vs Actual

Analyze variances.

Advantages

- Easy variance analysis
- Clear contribution breakdown

 Funnel Charts

Purpose

Displays stages in a process.

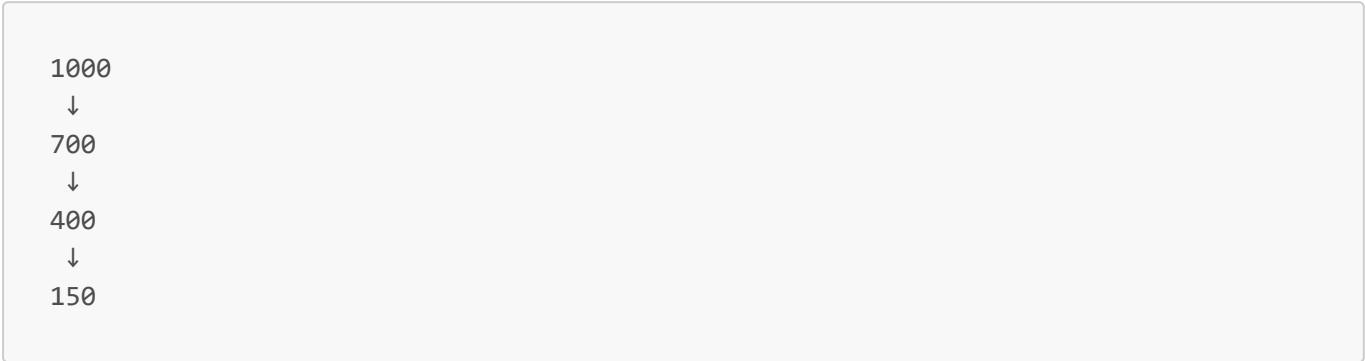
Example

Sales Pipeline

Stage	Leads
Leads	1000
Qualified	700
Proposal	400

Stage	Leads
Won	150

Visualization Concept



Business Use Cases

Sales Funnel

Lead conversion analysis.

Recruitment Funnel

Candidate selection process.

Marketing Funnel

Customer acquisition analysis.

Benefits

Identifies where drop-offs occur.

5 Scatter Charts

Purpose

Shows relationships between variables.

Example

Advertising Spend vs Revenue

Components

X-Axis

Advertising Spend

Y-Axis

Revenue

Bubble Size (Optional)

Profit

Business Use Cases

Correlation Analysis

Revenue vs Marketing Spend

Customer Analysis

Age vs Spending

Product Analysis

Price vs Sales

Interpretation

Positive Trend:

Higher Spend
→ Higher Revenue



Bubble Charts

Purpose

Enhanced scatter chart with a third dimension.

Example

Product	Revenue	Profit	Market Share
---------	---------	--------	--------------

Axes

X-Axis

Revenue

Y-Axis

Profit

Bubble Size

Market Share

Use Cases

Portfolio Analysis

Competitive Analysis

Market Segmentation

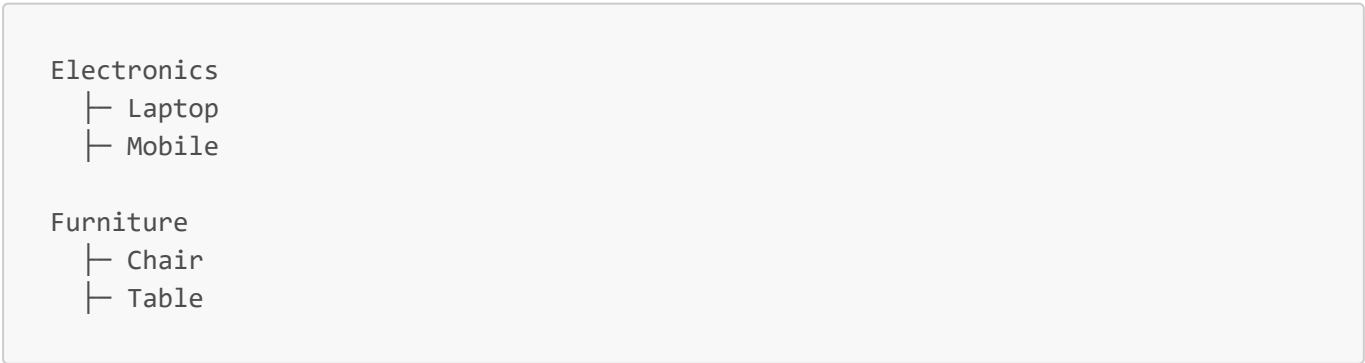
7 Treemaps

Purpose

Display hierarchical data using rectangles.

Example

Revenue by Category



Advantages

Shows:

- Proportions
 - Hierarchies
 - Category Contribution
-

Business Use Cases

Product Analysis

Revenue by category.

Expense Analysis

Cost by department.

Inventory Analysis

Stock by category.



Gauges

Purpose

Track progress toward a target.

Components

Current Value

Current performance.

Target Value

Expected performance.

Scale

Performance range.

Example

Sales Target

Current = ₹850K
Target = ₹1M

Business Use Cases

- Sales Targets
- Revenue Targets
- Productivity Tracking

Limitations

Avoid using too many gauges.

 Ribbon Charts

Purpose

Show ranking changes over time.

Example

Product Rankings

Month	Rank
Jan	Product A
Feb	Product B
Mar	Product A

Benefits

Shows:

- Rank Changes
- Market Leadership
- Competitive Position

Business Use Cases

Product Ranking

Brand Ranking

Sales Ranking

10 Decomposition Tree

Purpose

Perform root-cause analysis.

What Makes It Powerful?

Allows users to drill into contributing factors dynamically.

Example

Revenue

```
Revenue
  ↓
Region
  ↓
Product Category
  ↓
Product
```

Business Questions

Why did revenue decline?

Which region contributed most?

Benefits

- Interactive
- Dynamic
- AI-assisted

Real-World Use Cases

Sales Analysis

Customer Analysis

Operational Analysis

Key Influencers Visual

Purpose

Identifies factors influencing a metric.

Example

Question:

What drives customer churn?

Possible Influencers

- Age
 - Region
 - Product Usage
 - Subscription Type
-

AI Analysis

Power BI automatically identifies statistically significant factors.

Use Cases

Customer Churn

Employee Attrition

Product Success

Sales Growth

1 2 Q&A Visual

Purpose

Allows users to ask questions using natural language.

Example

User Types:

Total Revenue by Region

Power BI automatically generates a visualization.

Advantages

- Easy for business users
 - No DAX required
 - Self-service analytics
-

1 3 Smart Narrative

Purpose

Automatically generates textual insights.

Example

Instead of reading charts:

Revenue increased by 12%
compared to last month.

Benefits

Executive Reporting

Quick Insights

Automated Commentary

1 4 Custom Visuals

What Are Custom Visuals?

Additional visuals available beyond built-in Power BI visuals.

Source

AppSource Marketplace.

Popular Custom Visuals

Sankey Diagram

Flow analysis.

Bullet Chart

Advanced KPI tracking.

Gantt Chart

Project management.

Word Cloud

Text analysis.

Benefits

Expanded analytical capabilities.

1 5 Analytical Reporting Techniques

Advanced visuals should support analytical thinking.

Variance Analysis

Use:

- Waterfall Chart

Process Analysis

Use:

- Funnel Chart

Correlation Analysis

Use:

- Scatter Chart

Hierarchical Analysis

Use:

- Treemap
- Decomposition Tree

Root Cause Analysis

Use:

- Key Influencers
- Decomposition Tree

16 Choosing the Right Advanced Visual

Business Question	Recommended Visual
What caused revenue changes?	Waterfall
Where are leads dropping off?	Funnel
Is there a correlation?	Scatter
How do categories contribute?	Treemap
Are we meeting targets?	Gauge
How do rankings change?	Ribbon
Why did a KPI change?	Decomposition Tree
What drives a metric?	Key Influencers

17 Best Practices

Match Visual to Question

Always start with business objectives.

Limit Advanced Visuals

Use only when necessary.

Keep Design Simple

Avoid overwhelming users.

Use Tooltips

Provide additional context.

Combine with KPIs

Advanced visuals should complement key metrics.

1**8**

Common Mistakes

Using Advanced Visuals for Everything

Complexity doesn't equal value.

Overcrowding Dashboards

Too many visuals reduce clarity.

Ignoring User Audience

Executives often prefer simplicity.

Misinterpreting Correlation

Correlation does not imply causation.

Overusing Gauges

Consumes dashboard space.

Key Terminologies

Term	Definition
Waterfall Chart	Variance analysis visual
Funnel Chart	Process stage analysis
Scatter Chart	Relationship analysis
Bubble Chart	Scatter chart with size dimension
Treemap	Hierarchical proportion visualization
Gauge	KPI target tracking visual
Ribbon Chart	Ranking trend visualization
Decomposition Tree	Root cause analysis visual
Key Influencers	AI-driven factor analysis
Smart Narrative	Automated insight generation

Practice Exercises

Exercise 1

Create a Waterfall Chart showing:

- Revenue
 - Discounts
 - Returns
 - Net Revenue
-

Exercise 2

Build a Sales Funnel.

Stages:

- Leads
 - Qualified Leads
 - Opportunities
 - Closed Deals
-

Exercise 3

Create a Scatter Chart showing:

Marketing Spend vs Revenue

Exercise 4

Build a Treemap for:

Revenue by Product Category

Exercise 5

Use Key Influencers to analyze customer churn.



Mini Project

Advanced Sales Analytics Dashboard

KPI Section

- Revenue
 - Profit
 - Growth %
-

Advanced Visuals

Waterfall Chart

Revenue Variance

Funnel Chart

Sales Pipeline

Scatter Chart

Profit vs Revenue

Treemap

Category Contribution

Decomposition Tree

Revenue Analysis

Key Influencers

Sales Drivers

Objective

Build an advanced analytical dashboard capable of supporting executive decision-making.



Module Summary

In this module, you learned:

- ✓ Waterfall Charts
- ✓ Funnel Charts
- ✓ Scatter Charts
- ✓ Bubble Charts
- ✓ Treemaps
- ✓ Gauges
- ✓ Ribbon Charts
- ✓ Decomposition Trees
- ✓ Key Influencers
- ✓ Q&A Visuals
- ✓ Smart Narratives
- ✓ Custom Visuals
- ✓ Analytical Reporting Techniques

Advanced visualizations enable deeper analysis, root-cause investigation, and richer storytelling. They are especially valuable in executive dashboards, operational reporting, customer analytics, and business performance management.



Next Module

Module 15: Power BI Service & Cloud Reporting

Topics Covered:

- Introduction to Power BI Service
- Workspaces
- Publishing Reports
- Dashboards in Power BI Service
- Apps
- Data Refresh
- Sharing & Collaboration
- Security & Permissions
- Deployment Pipelines
- Governance & Administration